

Health Risk Lens

An explainable, rule-based screening dashboard that estimates risk for three chronic conditions, accounts for every point it awards, and simulates the effect of a change in behaviour.

Brief	Personalised Health-Risk Dashboard (problem statement #3)
Submitted by	
Document version	1.0
Date	10 September 2026
Implementation	Single-page web application · 3,755 lines · 13 UI components · no backend
Status	Feature-complete; 525 scoring assertions and a server-render smoke test passing

This is not a diagnostic tool. Every result it produces is a screening-level educational estimate generated by predefined scoring rules. The application does not diagnose, and it does not recommend any medicine or treatment. All profiles used in this report and in the application are synthetic. Any real health decision requires a qualified healthcare professional.

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1 Problem Statement

Chronic conditions such as type-2 diabetes, hypertension and cardiovascular disease develop over years and are substantially driven by modifiable factors — weight, physical activity, smoking and diet. Validated screening instruments for all three have existed in the clinical literature for two decades. The gap is not the mathematics; it is access and comprehension. The instruments live in journal appendices and clinical software, and a person who wants to understand their own standing has little between a full clinical consultation and an unsourced internet quiz.

Two failures are common in the tools that try to close that gap. The first is **opacity**: a tool returns "your risk is high" with no account of which inputs produced the result or how strongly, which makes the number impossible to trust and impossible to act on. The second is **fatalism**: a tool reports a static figure without showing that the figure responds to change, so a user with an elevated result learns only that they should be worried.

A third problem is specific to machine-learned health tools. A model trained on population data can be accurate in aggregate while being unable to explain any individual prediction, and in a clinical context an unexplainable output is difficult to act on, difficult to audit, and difficult to defend.

The problem this project addresses is therefore threefold:

- Estimate risk for multiple conditions from information a person can supply without laboratory work.
- Account for the estimate completely — every point, the rule that awarded it, and the published instrument that rule came from.
- Show what would change if the person changed something they can actually change, and be honest about what they cannot.

2 Proposed Solution

Health Risk Lens is a single-page web application that scores three conditions independently from thirteen items of basic health and lifestyle information, explains each score down to the individual rule row, and provides a simulator that recalculates all three scores in real time as modifiable factors are adjusted.

The design rests on three decisions, each taken deliberately against a more conventional alternative.

2.1 Rules, not machine learning

Scoring is performed by three point tables held as JSON and adapted from published instruments. There is no model, no training data and no inference. This is a deliberate trade: the system gives up any possibility of learning population-specific patterns in exchange for an explanation that is exact rather than approximate. When the application states that family history contributed five points, that is not an

estimate of feature importance — it is the arithmetic, and the rule row that produced it can be displayed alongside.

2.2 Three instruments, not one score

The three conditions are scored by three separate instruments with different factors, different weights and different cut-points. They are deliberately not normalised onto a shared scale, and the interface draws each condition's own band boundaries rather than a tidy shared split. A single blended "health score" would have been simpler to present and would have misrepresented every instrument it averaged.

2.3 Simulation as the same function, not a second one

Scoring is a pure function of a plain profile object. Simulation therefore requires no additional scoring logic: it is the identical function applied to a patched copy of the profile. Because both results come from the same point tables, the difference between them is the literal arithmetic, and the explanation of a simulated change cannot drift from the change itself.

Design constraint honoured throughout. The application is screening-level and educational. It does not diagnose, it recommends no treatment or medicine, and every disclaimer is persistent rather than a dismissible modal.

3 Features

Feature	Description
Three independent risk estimates	Type-2 diabetes, hypertension and cardiovascular disease, each scored by its own instrument and returning its own Low / Moderate / High band.
Four demo personas	One click replaces the whole profile and recalculates every score, chart and explanation on screen.
Health and lifestyle data entry	Thirteen fields with live derivation of BMI, waist circumference and blood-pressure category.
Instant recalculation	No submit button and no debounce; scoring is a few dozen table lookups per keystroke.
What-if simulator	Four modifiable levers plus three one-click scenarios; the original profile is never mutated.
Contributing-factor breakdown	Every factor that scored, its value, the points it awarded and the maximum it could have awarded, ranked, with the top factor flagged.
Data provenance drawer	Band thresholds drawn to scale and the complete rule table for every factor, with the row that fired ticked.
Comparison chart	The three results side by side, each column carrying its own band boundaries as background zones.
Simulation attribution	A row-by-row account of which rules changed, from what to what, and the resulting point delta per condition.
PDF export	A consultation-style summary document produced from a dedicated print component.
Accessibility provisions	Every band pairs colour with a glyph and a word; tabular figures prevent layout shift; sliders are keyboard-operable; reduced-motion preference is respected.

Table 1 — Feature summary.

4 System Architecture

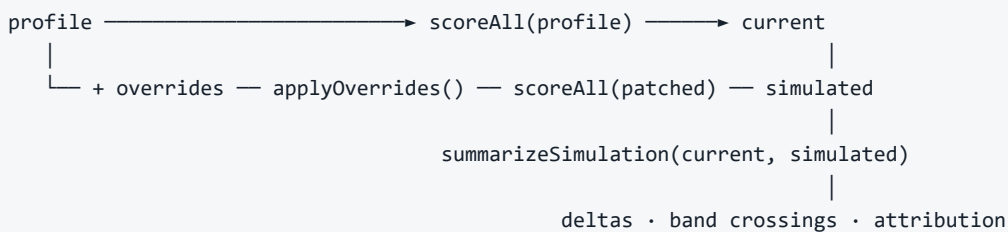
The application has no backend, no database, no network calls, no router and no state-management library. It is a single-page React application in which the rule tables are static JSON and all scoring is synchronous. This is an architectural choice with a specific benefit: a demonstration with no server dependency cannot fail because of connectivity, and the entire computation is inspectable in the browser.

4.1 Source layout

```
src/
  data/rules/*.json      the three point tables – the substance of the system
  data/factors.js        form metadata and which factors are modifiable
  data/demoProfiles.js   the four demo personas
  lib/scoring/           derive · scoreCondition · scoreAll · bands · conditions
  lib/simulation/        applyOverrides · summarizeSimulation · presets
  hooks/                 useRiskResults · useCountUp
  components/            layout · form · results · whatif · personas · print
  pages/                 Landing · Dashboard
  scripts/
    verify-scoring.mjs    525 assertions, plain Node, no test framework
    ssr-smoke.jsx         server-renders every screen and simulation state
```

4.2 Data flow

State consists of exactly two values held in the dashboard: the `profile` object and a sparse `overrides` patch. Everything else is derived on each render.



Because `overrides` is held separately and applied to a copy, the user's own data is never mutated. "Reset to my data" is not a re-fetch or an undo stack; it is the assignment of an empty patch.

4.3 The scoring engine

A single generic evaluator, `scoreCondition.js`, scores all three conditions. The difference between diabetes, hypertension and cardiovascular disease lives entirely in JSON: each rule file declares its factors, the rows each factor can match, the points each row awards, the band thresholds and the source citation. Adding a fourth condition requires a fourth JSON file and no new scoring code.

Bands are computed from the **raw** point total and never from the 0–100 display index, so the number shown on screen and the category beside it cannot disagree because of rounding.

4.4 Verification strategy

The project carries two executable checks rather than a conventional unit-test suite, both plain Node scripts with no test-framework dependency:

- `npm run verify` — **525 assertions** covering derived metrics and their guards, band boundaries including the exact BMI values 25.0 and 30.0, all four personas and their metadata, every step of the demonstration sequence, override hygiene (no-op overrides are dropped and the profile is never

mutated), the simulation summary including changes that make results worse and the empty difference, and the provenance invariants described in section 9.

- `npm run smoke` — server-renders both pages and six simulation scenarios: no overrides, a single change, the full sequence, a change that worsens the result, a no-op override, and an already-low profile. It fails if any component throws.

5 Technology Stack

Layer	Technology	Rationale
UI framework	React 18.3	Component model suits a screen where one state change must propagate to a dozen coordinated views.
Build tooling	Vite 6.0	Fast development server and an optimised production build with no configuration overhead.
Styling	Tailwind CSS 3.4	A token layer defined once in configuration — a four-step type scale, a spacing scale, three elevation tiers and a clinical palette — applied consistently across every component.
Charting	Recharts 2.13	Declarative SVG charting; custom background shapes allow each column to carry its own band thresholds.
Scoring	Plain JavaScript and JSON	No library. The rule tables are the domain model and are meant to be opened and read.
PDF generation	Browser print pipeline	Zero dependencies and true vector output; see section 10.
Verification	Node scripts	525 assertions and a server-render smoke test without a test-framework dependency.

Table 2 — Technology choices and the reason for each.

The full production dependency list is three packages: `react`, `react-dom` and `recharts`. Everything else is a build-time tool.

6 Risk Models Used

All three instruments are adapted from published sources, and every adaptation is stated in the interface itself rather than only in documentation.

Condition	Basis	Max	Low	Moderate	High
Type-2 diabetes	FINDRISC — Lindström & Tuomilehto, <i>Diabetes Care</i> , 2003	26	0–6	7–14	15–26
Hypertension	ACC/AHA 2017 blood-pressure categories with Framingham risk factors	27	0–7	8–15	16–27
Cardiovascular disease	Non-laboratory Framingham — Gaziano et al., <i>The Lancet</i> , 2008	38	0–11	12–22	23–38

Table 3 — Instruments and band thresholds, expressed in raw points.

6.1 Adaptations, stated plainly

- FINDRISC's five official categories are collapsed to three, as the brief requires.
- The cardiovascular points are re-scaled for this prototype and are **not** the published coefficient set. That model was chosen because it substitutes BMI for a lipid panel, and therefore requires no blood work — a direct fit for the no-laboratory-data constraint.
- Physical activity uses a single 150 min/week cut-point across all three models for consistency, following the WHO adult guideline.
- One family-history field feeds all three conditions rather than three separate fields.
- The source instruments are sex-binary. "Prefer not to say" resolves to the female reference table, which uses the lower waist thresholds.

6.2 Cross-condition coupling

A High diabetes result contributes three points to the cardiovascular score, because type-2 diabetes is itself an established cardiovascular risk factor. This makes the evaluation order load-bearing: diabetes is scored before cardiovascular disease. The coupling is non-circular by construction — cardiovascular risk never feeds back into the diabetes score — and it is surfaced in the interface as a labelled contributing factor rather than hidden inside the arithmetic.

The three sets of thresholds are deliberately **not** forced onto identical cut-points. Each condition draws its own band boundaries wherever it is displayed. Normalising them would have produced a tidier chart and a less accurate one.

7 Demo Personas

Four synthetic profiles ship with the application. The set is chosen so that the three models visibly **disagree** with one another: if every persona returned the same band across all three conditions, an observer could reasonably suspect a single hidden score presented three times. All four are verified by the assertion suite.

Persona	Profile	Diabetes	Hypertension	Cardiovascular
Healthy College Student	20 F · BMI 21.8 · never smoked · 120 min/week	8 Low	7 Low	5 Low
Busy Office Worker	38 M · BMI 28.4 · desk job · poor diet	38 Moderate	44 Moderate	24 Low
High-Risk Adult	58 M · BMI 30.1 · current smoker · sedentary	65 High	70 High	74 High
Senior Citizen	68 F · BMI 31.6 · prior high glucose	92 High	56 Moderate	53 Moderate

Table 4 — The four demo personas and their scores on the 0–100 display index. Every profile is fictional.

The office worker is the informative case: Moderate for diabetes and blood pressure but **Low** for cardiovascular risk, because he is young and has never smoked. The senior citizen is the complementary case — a high result driven largely by age and a previously recorded high glucose reading, neither of which any lifestyle change can alter.

8 What-If Simulator

The simulator is the feature the application is designed around. It exposes four levers — **smoking status**, **weight**, **physical activity** and **fruit and vegetable intake** — plus three one-click scenarios and a combined "apply all three". Systolic blood pressure is available under an Advanced disclosure, labelled as an outcome measure rather than a lever.

8.1 Only what a person can move

Age, sex, family history and previously recorded high blood glucose are displayed **locked**. They contribute to the score, but no lifestyle change alters them, and a simulator that let a user pretend to be thirty years younger would be a toy. Showing them as fixed is part of the argument the tool is making.

8.2 Two modelling choices surfaced rather than hidden

- **Quitting moves a user to *former smoker*, not *never*.** Risk remains elevated for years after cessation, and the point table reflects that. The interface states this next to the control.
- **Waist circumference tracks weight** at approximately 0.8 cm per kilogram. The diabetes instrument scores waist separately from BMI, so holding it constant would understate the effect of weight change. It is labelled as an estimate.

8.3 The demonstration sequence

The high-risk persona is numerically frozen so that the following sequence is reproducible. It is asserted step by step by the verification suite.

Step	Diabetes	Hypertension	Cardiovascular
Starting profile	65 High	70 High	74 High
Quit smoking	65 — no change	67 High	63 High
+ 150 min/week activity	58 High	59 High	55 → Moderate
+ lose 6 kg	46 → Moderate	56 → Moderate	42 Moderate

Table 5 — The demonstration sequence, asserted step by step by `npm run verify`.

The first step is deliberately included: smoking does not feed the diabetes model, so the diabetes card reports "no change to this risk" and explains why. A tool that moved every number whenever any input changed would be easier to demonstrate and would be lying. The third step crosses the two remaining band boundaries and simultaneously switches off the diabetes-to-cardiovascular coupling described in section 6.2.

9 Data Provenance and Explainability

Each contributing-factors card carries a **"How was this calculated?"** disclosure containing three things:

1. **The band thresholds** for that instrument, drawn to scale in proportion to the real point ranges, with a marker at the current raw score.
2. **Every rule row** each factor could have matched, with the row that fired ticked — so the reader sees not only what scored, but what the alternatives were and where the boundaries sit.
3. **The source instrument** and the adaptations made to it.

9.1 Why the drawer cannot lie

The rule rows are emitted by the scoring engine itself as part of each contribution record, not recomputed by the interface. The drawer is therefore a pure renderer: it has no independent arithmetic and cannot drift from the score displayed above it.

This property is asserted rather than assumed. For all **104 factor evaluations** across every persona, the verification suite checks that **exactly one row is active** and that **the active row's points equal the score that factor contributed**. The reasoning behind this test is simple: a provenance drawer that quietly disagreed with its own scoring engine would look, to any observer, exactly like a correct one.

9.2 Explanation of change

When the simulator is active, the interface additionally presents a row-level attribution of the difference — for example `smoking 6 → 2, -4`. Because both the current and simulated results are produced by the same point tables, this difference is the literal arithmetic and not an approximation of feature importance. The verification suite asserts that the per-factor deltas sum exactly to the reported score delta.

10 PDF Export

The "Export summary" action produces a consultation-style document containing a masthead, a profile table, a results table, the simulated changes when the simulator is active, and the disclaimer. It is a purpose-built component rendered for paper — not the dashboard with pieces hidden by a print stylesheet.

The implementation invokes the browser's own print pipeline against that component. This was chosen over `html2canvas` / `html2pdf` and `jsPDF` deliberately, for two reasons. First, it adds **zero dependencies**. Second, the output is real vector text — sharp at any zoom, selectable, searchable, and tens of kilobytes rather than a multi-megabyte rasterised screenshot. A canvas-based approach would additionally have mishandled the gradient callout, the translucent header and the sticky results strip.

Known limitation. Because this uses the browser print dialog, the user selects "Save as PDF" — two interactions rather than a silent download. A library such as `jsPDF` is the upgrade path if a true one-click download is required, at the cost of the dependency and the vector-text quality.

11 Screenshots

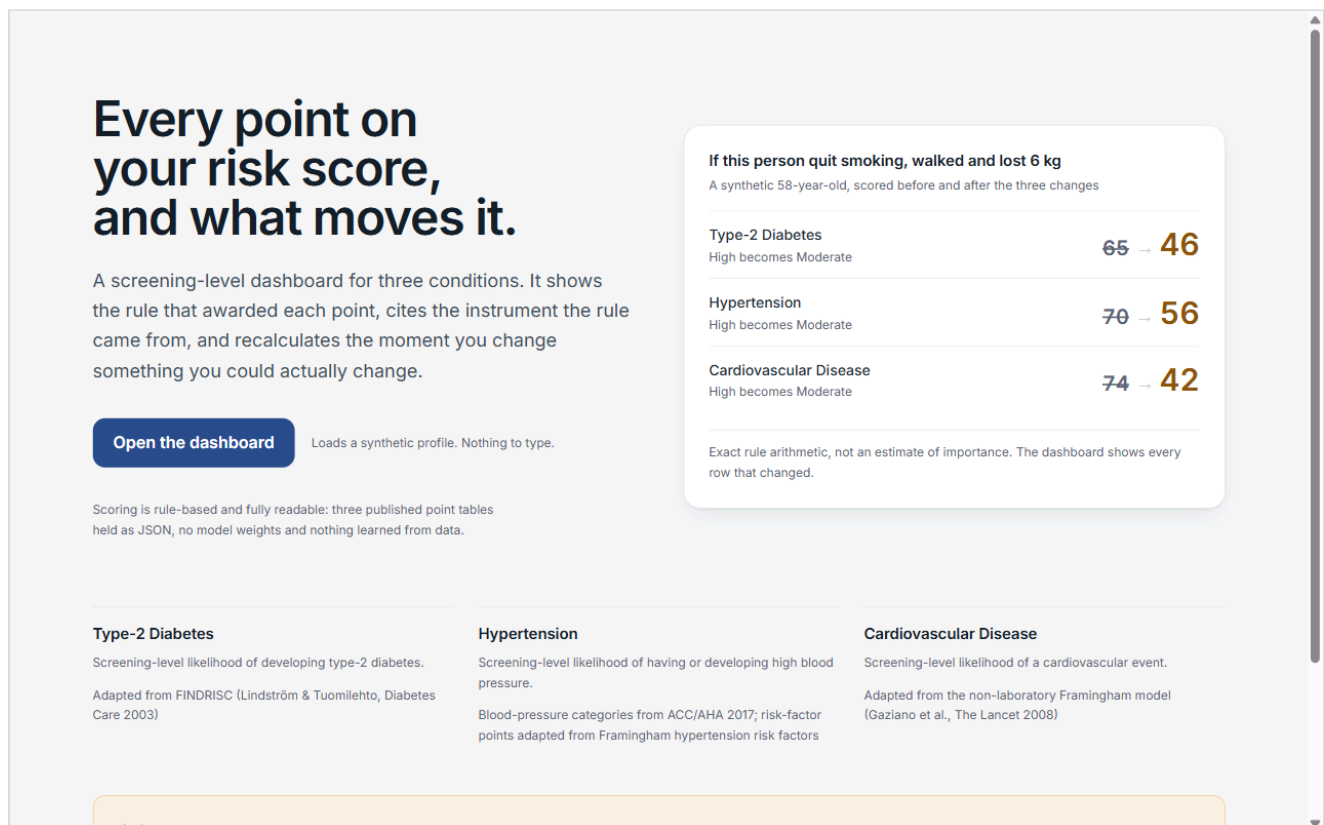


Figure 1 — Landing page. The panel on the right is not marketing copy: it is the real scoring engine run twice on the high-risk persona, once as-is and once with the three lifestyle scenarios applied. If a rule table changes, this panel changes with it.

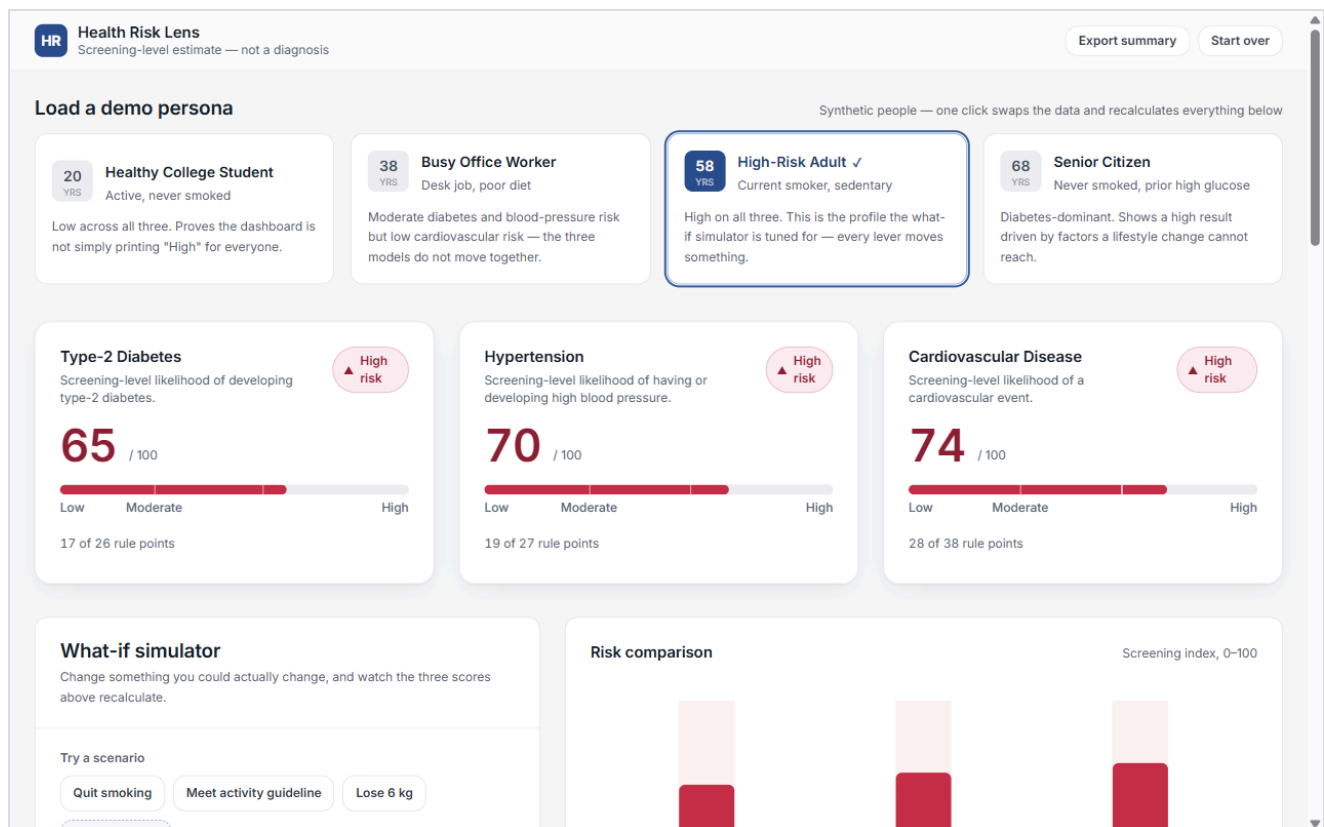


Figure 2 — Dashboard at rest. The persona selector, all three scores with their own band tracks, and the entry point to the simulator, on a single screen at 1280×800 with no scrolling required.

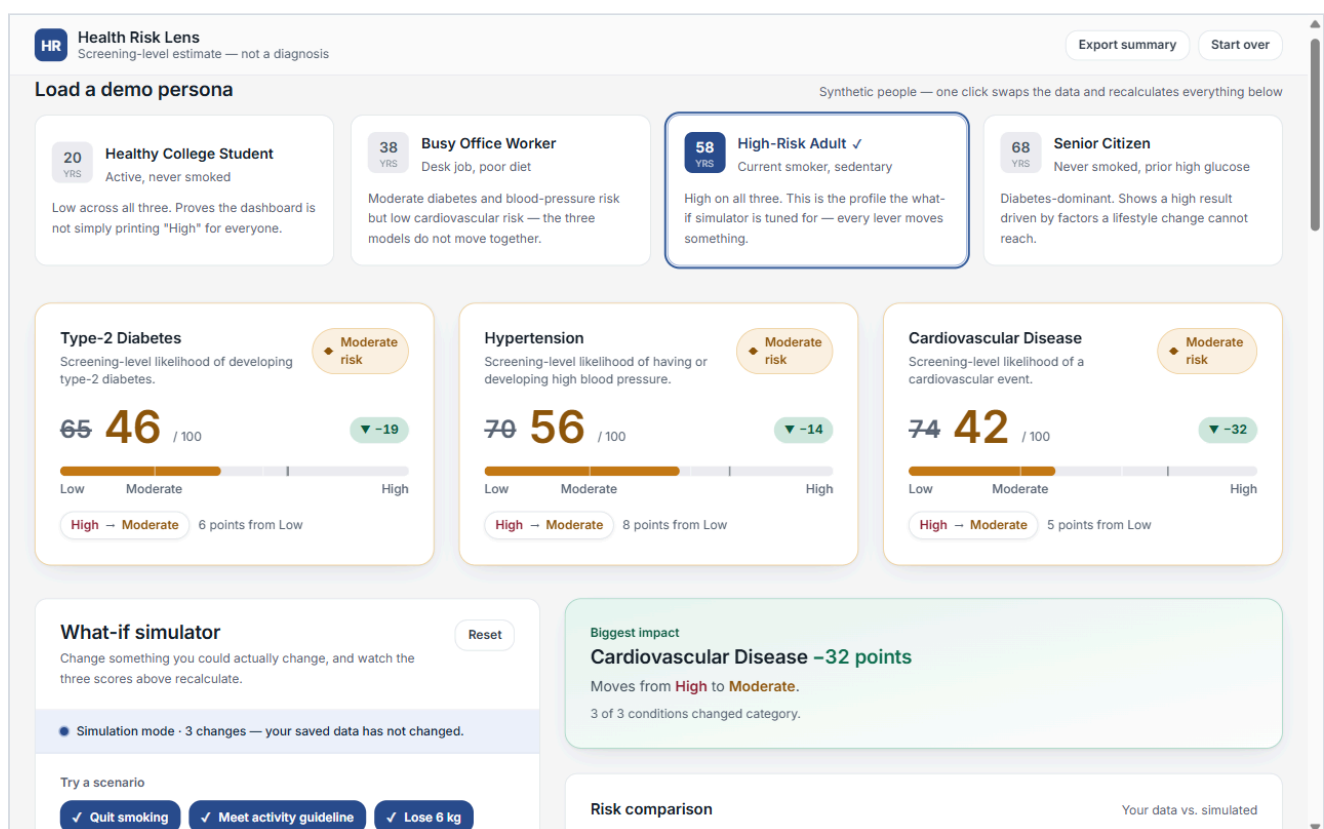


Figure 3 — Simulation mode. Three lifestyle changes applied. Each card shows the previous value struck through, the new value, the delta, and the band transition; the callout identifies the largest single effect. The original profile is unchanged.

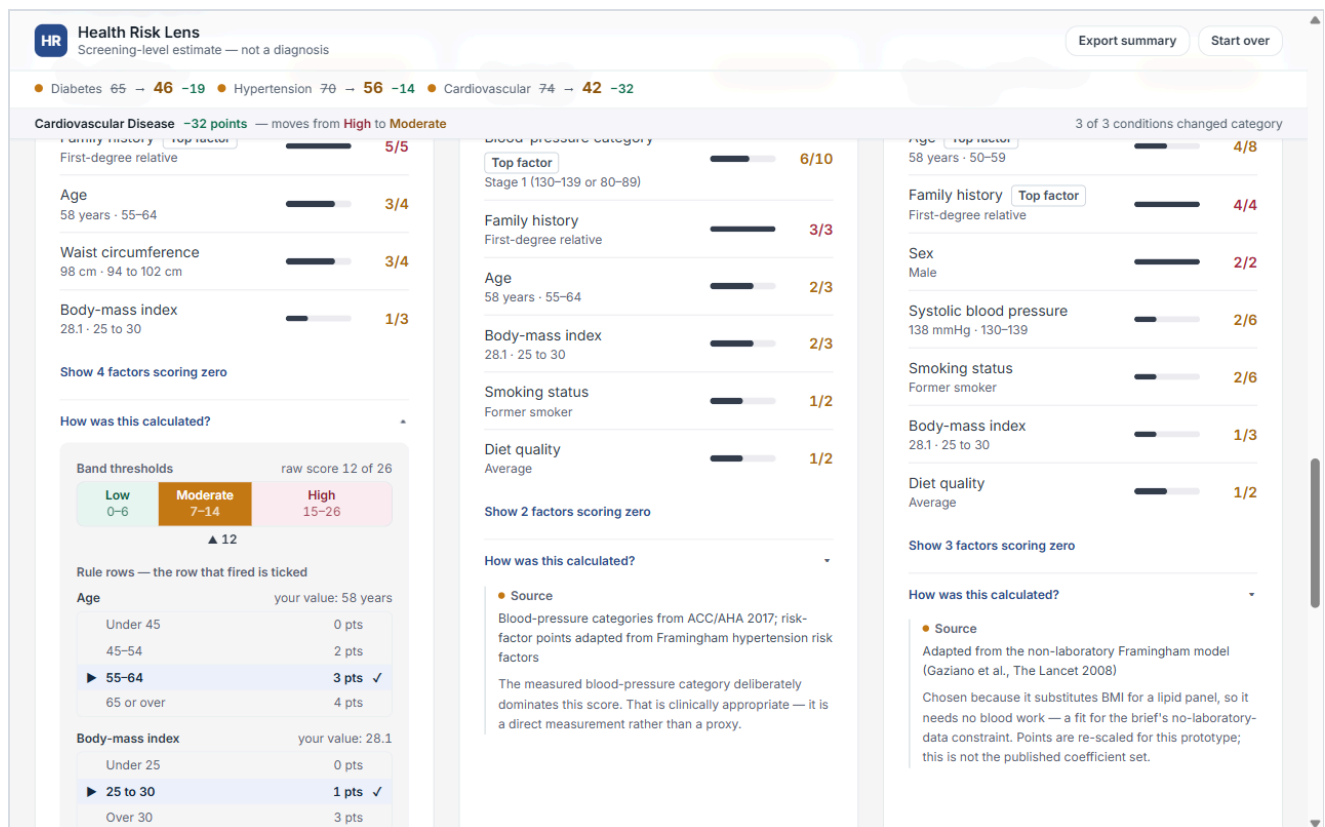


Figure 4 — Provenance drawer. Band thresholds drawn to scale with a marker at the raw score, followed by every rule row each factor could have matched, with the row that fired marked. These rows come from the scoring engine, not from the interface.

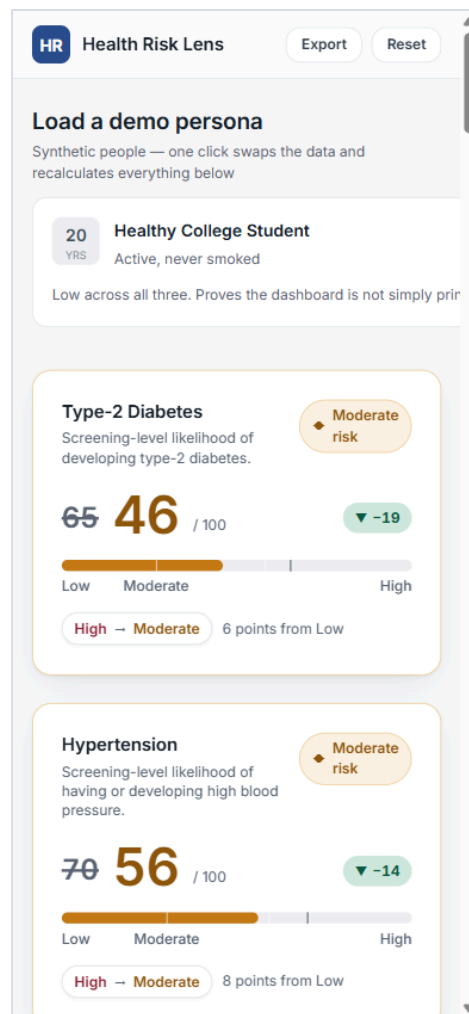


Figure 5 — Narrow viewport (390 px). *The layout reflows to a single column; scores, bands, deltas and transitions all remain legible.*

12 Future Scope

The following extensions are ordered by the ratio of value to architectural disruption. The first group requires no change to the existing design.

12.1 Near-term

- **One-click PDF download.** Replace the browser print dialog with a generated document, accepting the dependency cost described in section 10.
- **Additional conditions.** Chronic kidney disease and metabolic syndrome both have non-laboratory screening instruments. Each would require one JSON rule file and no new scoring code.
- **Profile persistence.** Local storage for a user's own profile, so a return visit does not begin from a demo persona. No backend required.
- **Formal accessibility audit.** The application already pairs colour with glyph and text, respects reduced motion and supports keyboard operation of every control; a formal WCAG 2.2 AA audit would confirm and complete this.

12.2 Medium-term

- **Laboratory-value variants.** Where a user has a recent lipid panel or HbA1c result, the full Framingham and clinical diabetes instruments become available and are more accurate. The rule-file architecture accommodates a second variant per condition.
- **Longitudinal tracking.** Storing dated profiles would turn the simulator's hypothetical into a record of actual change over time — the single most motivating feature the current design lacks.
- **Localisation.** The instruments are used internationally; the interface strings and the rule labels are already separated from the scoring logic.
- **Clinician view.** A denser presentation showing all three rule tables at once, intended for use during a consultation rather than by the patient alone.

12.3 Longer-term and out of scope for a prototype

- **Validation against real cohort data.** The adaptations in section 6.1 — particularly the re-scaled cardiovascular points — would need to be checked against outcome data before any claim of clinical accuracy could be made. Until then the tool is screening-level and educational, and says so.
- **Clinical system integration.** Ingesting values from an electronic health record would remove data entry entirely, and would introduce regulatory obligations substantially beyond the current scope.

Deliberately not planned. Replacing the rule tables with a trained model. It would make the explainability guarantees in section 9 impossible to maintain, and those guarantees are the point of the project.

13 Conclusion

Health Risk Lens demonstrates that a health-risk tool can be simultaneously useful and completely accountable. Every number it displays can be traced to a rule row, that row to a point table, and that table to a published instrument — and the interface exposes that chain to the user rather than documenting it elsewhere.

The central technical decision was to make scoring a pure function of a plain profile object. That one choice produced the simulator at almost no additional cost, guaranteed that the explanation of a change is the arithmetic of the change rather than an approximation of it, and made the whole system verifiable by ordinary assertions — 525 of them, including the invariant that the provenance drawer must agree with the engine that produced its data.

The project's principal limitation is stated plainly rather than minimised: the instruments are adaptations, the cardiovascular point set is re-scaled and not the published coefficients, and no result has been validated against outcome data. The application is a screening-level educational tool. It does not diagnose, it recommends no treatment, and it directs the user to a qualified healthcare professional. Within those boundaries it does something the opaque alternatives cannot: it shows its work, and it shows what would change if the user did.

End of report. Source, rule tables and verification scripts accompany this document.